

***In-situ* structure of outer membrane protein A: a novel structure refinement protocol**

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The structure and function of proteins are directly influenced by their surrounding chemical environment. This is particularly relevant for membrane proteins, which naturally reside in complex lipid bilayer membranes. However, structural and functional characterization of membrane proteins are generally accomplished in model systems, such as detergent micelles, lipid nanodiscs, or liposomes. Here we used ultra-fast MAS solid-state NMR spectroscopy to determine the structure of an outer membrane protein, OmpA from *Klebsiella pneumoniae* (a Gram-negative species), embedded in the native bacterial outer membrane, a highly asymmetric bilayer. Determining the structure and properties in such complex environment required a novel refinement protocol, developed specifically for the complex bacterial membrane. The outer leaflet consists of lipopolysaccharides with massive glycan head groups, which have low mobility due to a strong hydrogen bonding network; such low mobility makes protein insertion problematic, and the dynamics of the inserted protein particularly slow in the large head group region. Therefore, any protein structure has little chance to undergo any structural change after insertion. However, NMR-based information (on dihedral angles and inter-nuclear distances) is compatible with a broad range of structures, with differences concentrated in the four extra-cellular loops, in contact with the glycan group. In order to reproduce the structural variability of the loops, we devised a procedure in 5 steps, including an MD simulation step of the protein in a POPC membrane, with imposition of strong NMR-based restraints, allowing to sample many NMR-compatible conformations. Once embedded in the bacterial membrane, NMR-compatible structures are stable and different from the structures determined by solution NMR on the same protein in detergent micelles. Protein dynamics is slow, beyond the microsecond time scale, in agreement with NMR relaxation data. Overall, our new protocol yields a more realistic interpretation of protein conformational dynamics and protein interactions with the native bacterial membrane environment.